

KAARINA YLTOINEN, Finland

WMO: 028280

Lat: 60.387N

Lon: 22.551E

Elev: 5

StdP: 101.26

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.8	(c) -17.5	(d) -23.1	(e) 0.5	(f) -20.4	(g) -19.9	(h) 0.6	(i) -16.9	(j) 6.5	(k) -5.6	(l) 5.7	(m) -5.2	(n) 1.1	(o) 30	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	27.2	19.4	25.1	18.7	23.2	17.4	21.1	25.0	19.7	23.6	18.6	21.9	1.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.7	(b) 14.4	(c) 23.0	(d) 18.5	(e) 13.3	(f) 21.3	(g) 17.2	(h) 12.3	(i) 20.1	(j) 61.0	(k) 25.1	(l) 56.5	(m) 23.6	(n) 52.6	(o) 21.9	(p) 25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.4	4.5	4.0		-22.8	29.0	4.9	2.6	-26.4	30.8	-29.3	32.4	-32.0	33.8	-35.6	35.7	
				DB												
				WB	-23.0	22.0	4.9	1.8	-26.6	23.3	-29.4	24.3	-32.2	25.4	-35.8	26.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-5.3	-4.0	-1.3	4.0	10.3	14.3	17.4	15.9	11.8	5.7	2.4	-1.4
	DBStd	8.67	5.86	5.71	4.33	3.12	4.10	3.13	2.97	2.68	3.04	3.54	4.11	5.14
	HDD10.0	2179	473	391	349	181	46	3	0	0	17	137	228	354
	HDD18.3	4587	732	624	607	429	251	128	54	83	197	391	478	612
	CDD10.0	675	0	0	0	2	55	131	229	184	70	4	0	0
	CDD18.3	41	0	0	0	0	1	6	25	9	0	0	0	0
	CDH23.3	402	0	0	0	0	26	71	241	63	1	0	0	0
	CDH26.7	75	0	0	0	0	2	8	53	12	0	0	0	0
Wind	WSAvg	1.7	1.7	1.6	1.7	1.6	1.8	1.6	1.7	1.5	1.6	1.7	1.9	2.0
Precipitation	PrecAvg	679	58	40	34	33	34	54	69	84	58	77	69	68
	PrecMax	890	107	93	88	57	70	95	129	180	147	182	157	149
	PrecMin	479	5	0	0	4	6	9	0	22	9	15	7	5
	PrecStd	93	27	23	17	15	19	21	40	36	34	40	34	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	6.0	10.0	18.7	26.1	27.8	30.4	28.3	21.7	14.7	10.4	8.5
		MCWB	3.8	3.3	5.4	10.4	17.2	19.1	20.4	20.6	16.9	12.2	9.6	7.5
	2%	DB	3.5	4.1	7.5	15.0	23.4	25.4	28.0	24.8	19.6	13.0	9.2	6.7
		MCWB	2.8	2.6	4.1	8.7	15.1	17.8	20.5	19.3	15.8	11.4	8.7	5.8
	5%	DB	2.4	2.8	5.5	12.4	21.2	23.0	26.0	22.9	18.2	12.0	8.1	5.4
		MCWB	1.9	2.3	2.9	7.5	14.3	16.1	20.0	17.9	15.0	11.0	7.7	4.8
	10%	DB	1.5	2.0	3.8	10.2	18.6	21.0	24.1	21.4	16.9	11.1	7.3	4.4
		MCWB	1.0	1.6	2.0	6.2	12.8	15.1	18.8	17.0	14.6	10.2	6.7	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.1	6.1	11.0	18.2	20.6	23.0	21.8	17.8	13.2	9.9	7.9
		MCDB	4.4	5.5	8.8	17.5	22.7	25.4	27.4	25.5	20.3	13.8	10.2	8.3
	2%	WB	3.0	3.0	4.5	9.2	16.5	19.0	21.9	20.2	16.8	12.1	8.9	6.0
		MCDB	3.4	3.5	6.9	13.7	21.8	24.0	25.8	23.7	18.7	12.5	9.2	6.5
	5%	WB	2.0	2.3	3.3	8.0	15.0	17.5	20.6	18.9	15.9	11.3	7.9	4.9
		MCDB	2.4	2.8	4.8	11.8	20.4	21.7	24.4	21.5	17.5	11.7	8.1	5.3
	10%	WB	1.2	1.6	2.5	6.6	13.2	16.1	19.3	18.0	15.0	10.3	6.9	3.9
		MCDB	1.5	2.0	3.5	9.6	18.1	19.9	22.9	20.5	16.4	11.0	7.2	4.3
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.0	7.7	9.4	10.8	9.7	10.1	9.5	8.4	6.7	4.1	4.7
		MCDBR	3.8	5.8	8.9	14.3	15.0	13.8	13.2	12.0	10.3	7.4	4.3	5.0
		MCWBR	3.6	5.0	6.0	8.6	8.1	7.4	6.7	6.9	7.0	5.9	4.1	4.6
	5% WB	MCDBR	3.6	5.2	7.6	12.7	14.1	12.0	11.2	10.0	8.7	6.0	4.3	4.8
		MCWBR	3.5	4.6	5.6	7.9	8.1	7.0	6.4	6.3	6.4	5.4	4.2	4.6
Clear-Sky Solar Irradiance	taub	0.276	0.313	0.334	0.373	0.368	0.361	0.389	0.379	0.350	0.333	0.303	0.386	
	taud	2.149	2.228	2.308	2.322	2.401	2.448	2.392	2.413	2.483	2.397	2.213	2.613	
	Ebn at Noon	471	670	789	823	858	870	832	810	775	652	434	326	
	Edh at Noon	42	69	88	105	104	101	105	95	75	59	40	29	
All-Sky Solar Radiation	RadAvg	0.24	0.81	2.31	3.88	5.29	5.55	5.22	4.11	2.53	1.10	0.31	0.15	
	RadStd	0.04	0.16	0.28	0.39	0.52	0.44	0.51	0.43	0.32	0.18	0.06	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A	N/A

Nomenclature: See separate page